■ PCM Audio Reference Cheat Sheet

Bit Depth	Sample Type	Value Range	Bytes / Sample	Mono (1 ch)	Stereo (2 ch)	5.1 (6 ch)	7.1 (8 ch)
8-bit	Unsigned Int	0 … 255	1 byte	1 b/frame	2	6	8
16-bit	Signed Int	–32,768 … 32,767	2 bytes	2 b/frame	4	12	16
24-bit	Signed Int	–8,388,608 … 8,388,607	3 bytes	3 b/frame	6	18	24
32-bit int	Signed Int	–2,147,483,648 … 2,147,483,647	4 bytes	4 b/frame	8	24	32
32-bit float	Float	–1.0 … +1.0	4 bytes	4 b/frame	8	24	32
64-bit float	Float	–1.0 … +1.0	8 bytes	8 b/frame	16	48	64

■ Quick Byte Rate Formula

Bytes/sec = SampleRate × Bytes per Frame

Format	Sample Rate	Bytes/sec	KB/sec
16-bit Mono @ 8 kHz	8,000 × 2 = 16,000	16,000	16 KB/s
16-bit Stereo @ 44.1k	44,100 × 4 = 176,400	176,400	≈172 KB/s
16-bit Stereo @ 48k	48,000 × 4 = 192,000	192,000	187.5 KB/s
32-bit Stereo @ 48k	48,000 × 8 = 384,000	384,000	375 KB/s
24-bit Stereo @ 96k	96,000 × 6 = 576,000	576,000	562.5 KB/s
32-bit Float 5.1 @ 48k	48,000 × 24 = 1,152,000	1,152,000	≈1.1 MB/s